

# MAX SONG

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## EDUCATION

### The University of Texas at Austin

December 2027

Bachelor of Science in Mechanical Engineering | *Design and Manufacturing Track*

GPA: 3.8

Relevant Coursework: Solids, Dynamics, Materials, Fluids, Computational Methods, Statics, Thermodynamics

## SKILLS

**CAD/CAE:** Solidworks (Assemblies, Drawings, BOM), Fusion 360 (CAM), Autodesk Inventor, Blender, Abaqus FEA, ANSYS, MS Office

**Analysis:** MATLAB, Python (OpenCV, Matplotlib, Pandas, NumPy), FEA (Static, Modal), GD&T, Tolerance Stacks, Structural Analysis

**Manufacturing:** CNC Mill, Manual Mill, Manual Lathe, Laser Cutting, SLA and FDM Printing, Soldering, Injection Molding, Table Saw

**Materials:** Carbon Fiber, Fiberglass, Basalt Fiber, Glass beads, Al 6061-T6, PLA/PETG/TPU/ABS/PA-CF, Elastomers, PMMA

## EXPERIENCE

### Contoro Robotics – Robotics Engineering Intern; Austin, TX

January – August 2026

- Engineered an end-of-line gripper-test stand to streamline manufacturing, integrating master IO Links with Festo manifolds and pressure transducers for pneumatic and vacuum validation, implementing a redesigned and manufactured rotational end-effector assembly stand validated via hand calculations and re-sourced the BOM to cut stand cost by 36%
- Created and released electromechanical assembly drawings and full BOMs in SolidWorks for entire robot assembly including junction boxes, power and data conduits, and sensor towers, designing retrofits to improve clearance and increase throughput
- Manufactured full-scale warehouse unloading robots and operated autonomous mobile robots at an Amazon fulfillment center

### Texas Rocket Engineering Lab; Austin, TX

September 2024 – Present

*Halcyon Mark 1 – Mechanical Structures Engineer; Composites Manufacturing Engineer*

- Led the development of a 48-inch length nosecone in SolidWorks, utilizing DFM principles for composite fiberglass manufacturing, using a 3D printed ABS vacuum-forming mold and troubleshooting Gigabot large format FDM printer for mechanical failures
- Fabricated flight-critical components (carbon fiber skirts and couplers, fiberglass raceway mounts) using hand layups and vacuum bagging techniques for Halcyon Mark 1 (a liquid bipropellant spaceshot rocket), passing 9600 lbf axial structural validation test

### Orbital Test Stands – Structures Engineer

- Spearheaded CAD model of 30-ft test stand structure for hot-fire engine testing with 30+ components (steel grate flooring, I-beams, brackets, anchors) in SolidWorks for Preliminary Design Review (PDR) enabling structural integration
- Conducted root-cause analysis, redesigning pressure transducer brackets with isolation improving sensor accuracy during testing
- Developed Excel ACI 318 bolt calculator achieving 150% load capacity while reducing bolt count with standard GSE safety factors

### Texas Inventionworks Makerspace at the University of Texas at Austin – Student Technician; Austin, TX

August 2024 – Present

- As woodshop lead, directed cross-functional team of 10+ student staff to develop preventive maintenance (PM) schedules, safety protocols and SOPs for 12+ machines (planer, jointer, table saw, router, bandsaw) increasing equipment uptime by over 20%
- Instructed 100+ students in hands-on curriculum covering manual mill, lathe, CNC, SLA/FDM 3D printers on machine shop team
- Led student engagement initiatives including project competitions like Pinewood Derby to improve student participation by 35%

### Architected Intelligent Matter Lab at the University of Houston – Materials Research Intern; Houston, TX

June – September 2025

- Developed layered casting method for thin-shell elastomer half-spheres using precision steel ball mold with laser-cut acrylic jig; performed modeling and layer thickness testing achieving  $\pm 0.2$  mm dimensional accuracy for acoustical metamaterial testing
- Improved Abaqus FEA modal simulation accuracy by 10% using Instron ASTM D412 tensile tests and OpenCV video extensometry
- Automated material characterization workflow using Python reducing analysis time by 60% enabling rapid design iterations

## PROJECTS

### Custom Macropad Keyboard – Electronics and Product Design Engineer

November 2024

- Designed complete custom assembly and PCB layout, performing component selection, circuit design, and tolerance analysis
- Manufactured prototypes through complete workflow including FDM/SLA 3D prints (PLA housing, PETG plate, resin keycaps), PCB hand soldering (switches, diodes, resistors), Python firmware programming, and electrical validation testing with multimeter

### Battle Bots Design Competition – Electronics and Weapons Engineer

October 2024

- Designed complete robot assembly featuring optimized single-tooth vertical spinner weapon with center of mass analysis and integrated electronics system including motors, ESC hand soldering, radio receiver integration, and controller configuration